

	1 Thread	2 Threads	4 Threads	8 Threads
getTransaction	17.40 secs	10.73 secs	8.40 secs	24.59 secs
getTransactionFromFile	19.85 secs	51.20 secs	92.81 secs	271.32 secs

1. In what scenarios did threads improve performance and why?

Additional threads improved performance when using the getTransaction function up to 4 total threads. This is because the 4 cores available are all being utilized to run one thread each.

2. In what scenarios did threads have a negative impact on performance and why?

At 8 total threads when using the getTransaction function, the performance was worse overall due to the limitation of the number of cores. Having fewer cores than threads causes more context switches to be needed to continue executing in parallel.

When using the getTransactionFromFile method, more threads always causes a degradation in performance. This is because filesystems utilize locking to maintain thread safety, meaning each thread must wait for the lock to be released to access the information.